Insurance

MGT 4074 Fintech & Crypto Tokens, Section A, Spring 2025

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Roadmap

1. In the News
2. Insurance Industry Facts
3. FinTech Impact on the Insurance Sector
4. Incumbent Responses

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Blackrock (world's largest index fund provider) on Tokenization

Tokenization is democratization

The world's money moves through plumbing built when trading floors still shouted orders and fax machines felt revolutionary.

Take the Society for Worldwide Interbank Financial Telecommunication (SWIFT). It's the system that underpins trillions of dollars in global transactions every day, and it works much like a relay race: Banks hand off instructions one by one, meticulously checking details at each step. That relay approach made sense in the 1970s, an analog era when the markets were much smaller and daily transactions were much fewer. But today, relying on SWIFT feels like routing emails through the postal office.

Tokenization changes all that. If SWIFT is the postal service, tokenization is email itself—assets move directly and instantly, sidestepping intermediaries.

What exactly is tokenization? It's turning real-world assets—stocks, bonds, real estate—into digital tokens tradable online. Each token certifies your ownership of a specific asset, much like a digital deed. Unlike traditional paper certificates, these tokens live securely on a blockchain, enabling instant buying, selling, and transferring without cumbersome paperwork or waiting periods.

Every stock, every bond, every fund—every asset—can be tokenized. If they are, it will revolutionize investing. Markets wouldn't need to close. Transactions that currently take days would clear in seconds. And billions of dollars currently immobilized by settlement delays could be reinvested immediately back into the economy, generating more growth.

Perhaps most importantly, tokenization makes investing much more democratic.

It can democratize access. Tokenization allows for fractional ownership. That means assets could be sliced into infinitely small pieces. This lowers one of the barriers to investing in valuable, previously inaccessible assets like private real estate and private equity.

It can democratize shareholder voting. When you own a stock, you have a right to vote on the company's shareholder proposals. Tokenization makes that easier because your ownership and voting rights are digitally tracked, allowing you to vote seamlessly and securely from anywhere.

It can democratize yield. Some investments produce much higher returns than others, but only big investors can get into them. One reason? Friction. Legal, operational, bureaucratic. Tokenization strips that away, allowing more people access to potentially higher returns.

One day, I expect tokenized funds will become as familiar to investors as ETFs—provided we crack one critical problem: **identity verification.**

Financial transactions demand rigorous identity checks. Apple Pay and credit cards handle identity verification effortlessly, billions of times a day. Trade venues like NYSE and MarketAxess manage to do the same for buying and selling securities. But tokenized assets won't run through those traditional channels, meaning we need a new digital identity verification system. It sounds complex, but India, the world's most populous country, has already done it. Today, over 90% of Indians can securely verify transactions directly from their smartphones.[60]

> **"Every stock, every bond, every fund—every asset—can be tokenized. If they are, it will revolutionize investing.**

State Farm Seeks Approval to Raise Premiums

State Farm Threatens to Abandon California If They Can't Raise Prices: 52% For Renters, 30% For Homeowners

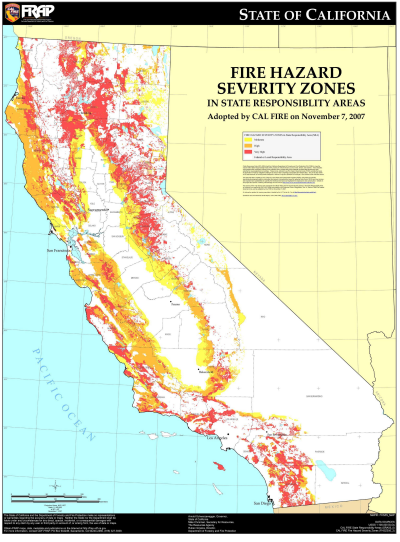
State Farm plans to cut ties with 72,000 California customers



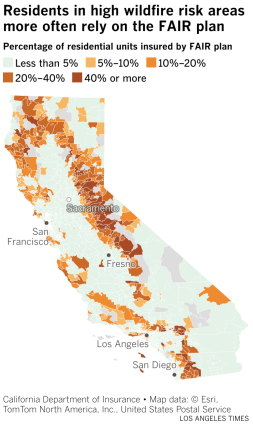
By **Niloy Chakrabarti**

Published 12 July 2024, 1:14 PM BST

California Fire Insurance is Strictly Regulated

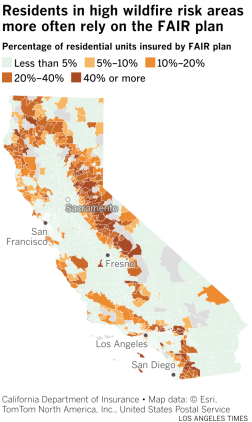


California's FAIR Plan



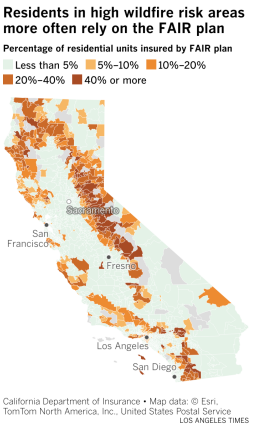
Many Californians are offered **zero** fire insurance plans

California's FAIR Plan



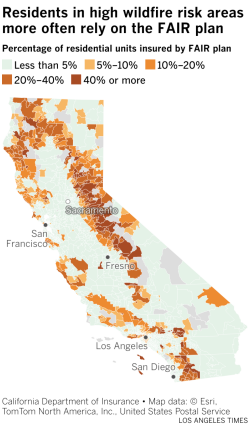
Many Californians are offered **zero** fire insurance plans, but are required to maintain fire insurance.

California's FAIR Plan



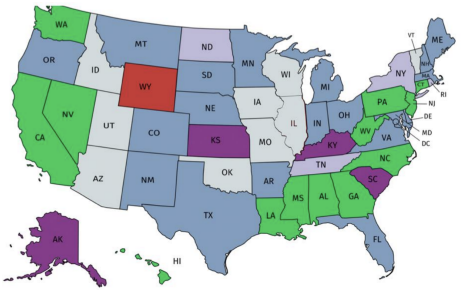
Many Californians are offered **zero** fire insurance plans, but are required to maintain fire insurance. The state-run FAIR plan is their only option

California's FAIR Plan



Many Californians are offered **zero** fire insurance plans, but are required to maintain fire insurance. The state-run FAIR plan is their only option, which is bankrupt and currently requiring a $500 billion bailout from taxpayers.

Legal Barriers to Innovation in Insurance Pricing



California is one of only many states to regulate insurance premiums, often to below actuarially-fair prices.

Roadmap

1 In the News

2 Insurance Industry Facts

3 FinTech Impact on the Insurance Sector

4 Incumbent Responses

Insurance Industry Basics

The global insurance industry is a giant

- The size of the insurance industry is measured by the dollar value of insurance premiums paid by customers, called **gross written premiums (GWPs)**.
- GWPs were estimated at US$5.7 trillion in 2020

An **insurance policy** is a contract between an insurance company and customer.

- The insurance company underwrites the policy, agreeing to pay out money when something bad happens (e.g., auto accident).
- The customer pays a one-time or regular premium for this insurance coverage.
- When a claim is paid out to a customer, we call it a settlement. Claims are the main expenses for insurers.
- Insurance creates value for society by transferring money from the lucky (low marginal utility) $\rightarrow$ to the unlucky (high marginal utility)

The Need for Regulation



The Need for Regulation

Market failures underlie the need for regulation in insurance:

- **Claim Adjudication**: who decides whether a real-world event meets the definition of an "insurable event" according to the contract?
  - The O.G analogue to the **oracle problem** in DeFi
  - Heavy reliance on local law enforcement and "independent" arbitrators

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  - Heavy reliance on local law enforcement and "independent" arbitrators
- **Market Power**: Insurers must be deeply capitalized and face huge economies of scale $\rightarrow$ lack of competition
  - Regulators often cap prices directly to prevent insurers from charging high markups
  - A more market-based approach is to cap the rate of return on insurers' capital
  - However, rate-of-return caps lead to under-investment

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  - However, rate-of-return caps lead to under-investment
- **Adverse Selection**: Health insurance will be a bad deal for people who know they are healthy!
  - Massachusetts and later Affordable Care Act ("Obamacare") sought to solve this with the "individual mandate" (everyone must buy insurance even when it's a bad deal for them), later repealed federally
  - All states mandate drivers hold auto insurance

Insurance Industry Sectors

Major insurance sectors are: life insurance (42% of GWPs); property and casualty (31%), and health insurance (27%).

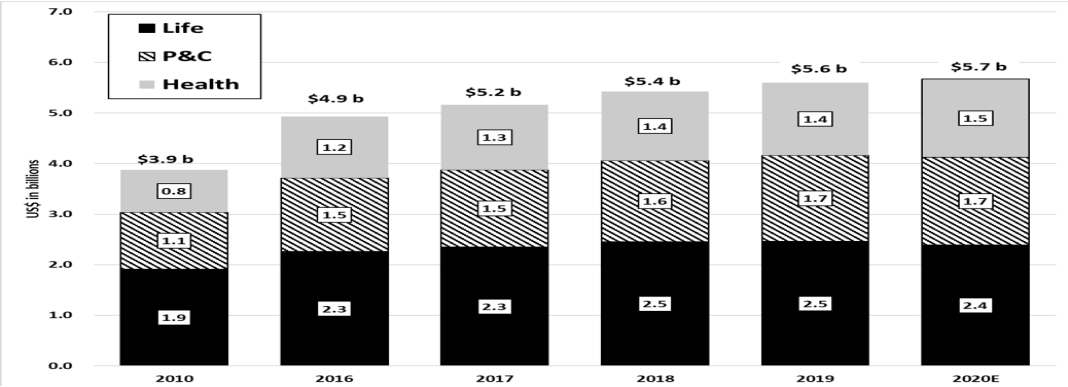


Figure: Gross Written Premiums in the Global Insurance Industry

Insurance Industry Facts and Pain Points

The insurance industry has two highly profitable B2B sectors:

- **Policy distribution and management**: insurance brokers selling and servicing policies written by the big insurance companies.
- **Reinsurance**: an insurance company buys insurance to protect against claims on their policies, transferring the risk to another party

The insurance industry is profitable but features many pain points for customers.

- Crude pricing (either by law or by antiquated practices)
- Inefficient Paper-Based Onboarding & Claims Processing
- Expensive Distribution Networks
- General Lack of Service

Insurance Industry Facts and Pain Points

Largest Pain Point: "Actuarially fair" pricing based on few observable characteristics $\rightarrow$ large mispricings for individuals

- Insurance products are generic and commoditized, with little to distinguish policies written by major insurers.
- e.g., car insurance for under 25-year old male drivers does not distinguish based on driving history

Complexity, regulation, and economies of scale are barriers to entry for insurtechs.

- No market for undiversifiable risks (e.g., forest fires in interior British Columbia, where either every tree or no tree will burn down $\rightarrow$ no scope for diversification)
- Underwriting does not lend itself to a P2P model; insurers need scale and portfolio diversification to spread the risk and earn a profit.

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Fintech Value Proposition

The Fintech Value Proposition in Insurances: FinTech companies can provide **personalization** with **lower overhead prices**, **faster claim processing**, and **better customer experience**.

Crude Pricing: broad, outdated risk models, leading to inaccurate pricing.

- AI & Data-Driven (driving behavior, health metrics, Internet-of-Things data) Models for personalized pricing
  - Usage-Based Insurance (UBI): Auto insurers use telematics to track driving habits and adjust premiums accordingly (e.g., Metromile, Root Insurance).
  - Real-time Risk Assessment: Insurers can dynamically price policies based on real-world data instead of static demographics.

Fintech Value Proposition

Inefficient (Manual) Paper-Based Onboarding & Claims Processing

- Digital Automation & Smart Contracts
  - AI-powered digital onboarding reduces the need for paperwork (e.g., selfie-based ID verification, e-KYC (know your customer)).
  - Blockchain & Smart Contracts automate claims processing, ensuring faster, transparent payouts (e.g., Lemonade's AI-driven instant claims).
  - Chatbots & AI Assistants handle customer queries and streamline documentation submission.

Strategy for Fintechs

Expensive Distribution Networks: costly human agents and brokers

- Direct-to-Consumer (D2C) Digital Insurance
  - InsurTech startups offer direct-to-consumer (D2C) digital insurance, cutting out intermediaries (e.g., Oscar Health, Hippo).
  - Embedded Insurance: Platforms like Amazon or Tesla bundle insurance at the point of sale, reducing distribution costs.
  - Comparison Platforms: AI-powered aggregators (e.g., Policygenius, CoverHound) help consumers find the best policy in seconds.

Strategy for Fintechs

General Lack of Service: poor service, complex policies, and delayed responses.

- AI-Driven Personalization & Instant Support
  - AI Chatbots & Virtual Assistants provide 24/7 customer support.
  - Data-Driven Personalization: AI suggests tailored policies based on customer behavior, improving engagement.
  - Microinsurance & On-Demand Coverage: InsurTechs offer flexible, short-term coverage (e.g., Trov, Bima for travel, gig workers).

Investment in Insurance Fintechs

Investment in Insurtech took off around 2015, later than other fintech sectors:

- Insurtechs raised $50 billion from 2018 to 2022
- CB Insights tracking 44 insurtech unicorns globally in 2022
- 20 insurtechs went public through U.S. IPOs in 2020
- Exits via mergers and acquisitions have skyrocketed

Investment in Insurance Fintechs

Mixed results for insurtechs, with some rising and others falling post-IPO.

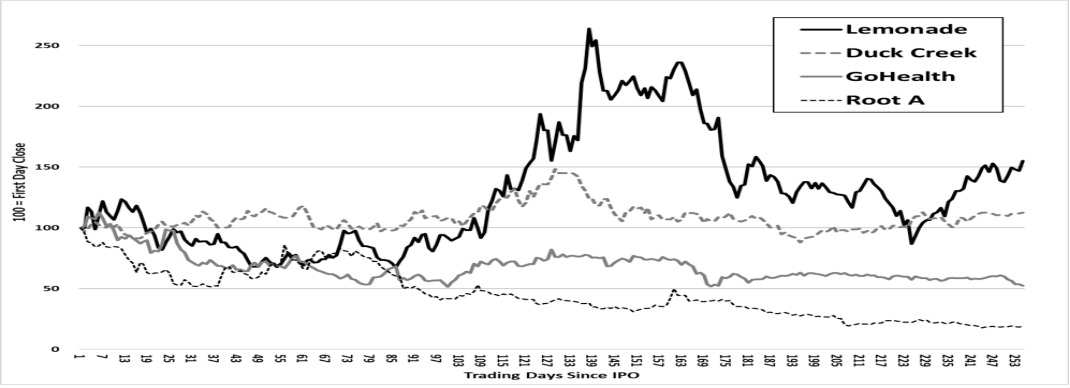


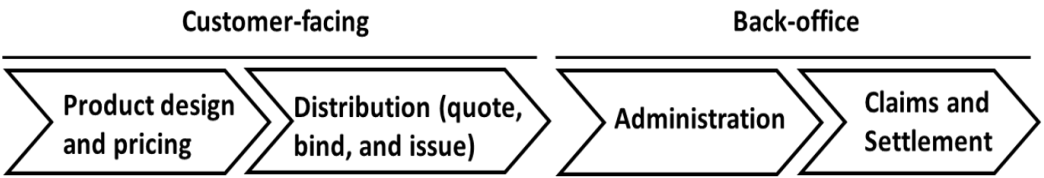
Figure: Insurtech Performance First Year Post-IPO

The Insurance Value Chain

Economies of scale and low "interest factor" limit competition in each segment

- Individualized pricing is useless when regulators fix prices and quantities
- Better actuarial prediction is useless when monopoly power means pricing is based on demand instead of risk

Insurtechs need to pick their spot on the value chain and do it better, cheaper, faster and more conveniently.



FinTech Example: Auto Insurance

Traditional insurance companies face difficulties in insurance pricing because of adverse selection and moral hazard due to asymmetric information.
Adverse Selection:

- High-risk individuals are more likely to buy insurance than low-risk individuals, but insurer companies cannot perfectly distinguish between them.
- **Consequence: Higher Average Premiums**. To cover the increased claims from high-risk customers, insurers raise premiums for everyone.

Traditional solution:

- Mandates & Regulations: Government-imposed insurance requirements (e.g., mandate auto insurance, the Affordable Care Act's individual mandate) prevent only high-risk individuals from buying coverage.
- Underwriting & Screening: group screening based on demographics (age, ZIP code, credit score)

FinTech Example: Auto Insurance

Traditional insurance companies face difficulties in insurance pricing because of adverse selection and moral hazard due to asymmetric information.

Moral Hazard:

- Insured individuals take more risks because they do not bear the full financial consequences of their actions.
- **Consequence: Higher Premiums or Deductibles**. To compensate for riskier behavior, insurers may increase premiums, introduce deductibles, or adjust coverage limits.

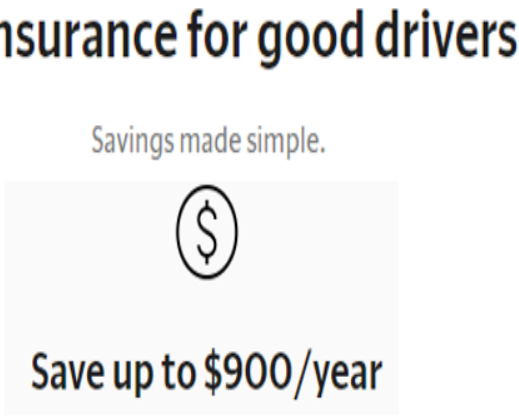
Traditional solution:

- Co-Payments & Deductibles: Customers share some financial risk, discouraging reckless behavior.
- Monitoring & Incentives: Insurers reward policyholders for responsible actions (e.g., safe drive incentive: premium drops after long-term no claim; health incentives: premium drops after long-term fitness tracking).

FinTech Example: Auto Insurance

Root offers better solutions to personalized pricing by taking advantage of technological advancement





FinTech Example: Root Auto Insurance

Root offers better solutions to personalized pricing by taking advantage of technological advancement

Solution to Adverse Selection: Mobile-Only Sign-Up & Telematics-Based Pricing

- Instead of relying on demographics (age, ZIP code, credit score), Root uses smartphone sensors (GPS, accelerometer, gyroscope) to track driving habits (Test-Drive Period of several weeks) before offering a quote.
- If a driver is too risky (aggressive acceleration, hard braking, phone use while driving, speeding), Root may deny coverage entirely—reducing adverse selection by filtering out high-risk drivers before they even enter the pool.

Solution to Moral Hazard: Continuous Monitoring & Behavior-Based Pricing

- Safer drivers get rewarded; Reckless driving (frequent speeding, sharp turns, distracted driving) results in higher premiums

FinTech Example: Auto Insurance

Root offers better solutions to personalized pricing by taking advantage of technological advancement

Personalized Price

- **Usage-based pricing**: besides driving habits, driving mileage per month also matters. Customers can pause coverage (in some cases) when not driving.
- **Fully Digital & Seamless Experience**: 100% app-based insurance underwriting & claims processing based on AI & automation, much faster than human-based paper work.
- **Cost efficiency**: Operates digitally without brick-and-mortar offices (agent commissions, and expensive infrastructure).

FinTech Example: Auto Insurance

Root faces several challenges:

- State Regulations: Some states (Michigan, New Hampshire, New Jersey, New York, North Carolina, Rhode Island, South Dakota, and Vermont) require traditional underwriting factors, limiting Root's data-driven model.
- Customer Complaints: Some users report inconsistent pricing or issues with claims, indicating its AI model is still in an early stage.
- Profitability Struggles – Usage-based insurance models can be hard to sustain at scale, especially when acquiring new customers, especially with early-stage AI model.

FinTech Example: Lemonade as All-around Insurance

Traditional Insurance Companies face conflicts of interest: Claim embellishment due to incomplete contract.

Incomplete contract: there is always some middle ground where one claim can be reasonably declined or accepted.

- Insurance companies benefit from declining claims that are unreasonable, weak, and out of scope.
- Insurance buyers benefit from embellishing the claims to increase the probability of acceptance.

Consequence:

- False decline: some reasonable claims are not accepted due to non-embellishment.
- False acceptance: some unreasonable claims are accepted due to embellishment.

FinTech Example: Lemonade as All-around Insurance

Traditional Insurance Companies face conflicts of interest: Claims embellishment due to incomplete contract.

Lemonade uses game theory to solve this claim embellishment.

- Fixed-fee ex-post: Lemonade charges a fixed 20% fee of unclaimed premiums;
- "Giveback" model ex-post: 80% unclaimed premiums are donated to charities that are chosen by insurance buyers ex-ante.

Consequence:

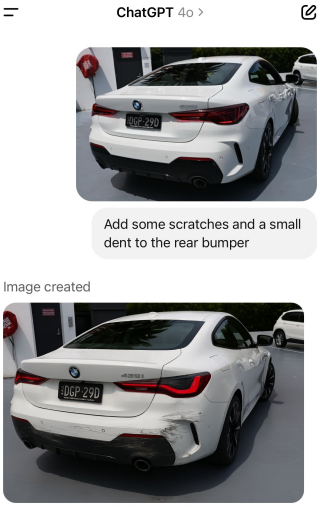
- Lemonade has less financial incentive to decline claims.
- Insurance buyers have less incentive to embellish claims: claiming unreasonable accidents only hard buyer-chosen charities.
- Both "false decline" and False acceptance" decline.

FinTech Example: Lemonade as All-around Insurance

Lemonade also uses technological advancement to reduce costs:

- Uses an AI chatbot (Maya) to process applications in 90 seconds or less.
- AI chatbot (Jim) processes 30% of claims instantly using fraud detection algorithms.
  - Average claims processing time as fast as 3 minutes for simple cases.
  - Note most human assistance in claim processing is to solve the unclear middle group issues. Eliminating "claim embellishment" goes a long way to reduce such costs.
- Uses blockchain (immutable) to prevent double dipping (file the same claim multiple times with different insurers) since other insurance companies can check.
- Recording: users must record a video statement explaining their loss when making a claim
- No brokers: direct-to-consumer model powered by AI, reducing operational costs.
- Dynamic pricing adjusts rates in real-time using machine learning

Insurance Fraud is an Eternal Cat-and-Mouse Game



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Incumbent Responses

1. In-House AI & Telematics-Based Products

- AI-Powered Risk Assessment & Dynamic Pricing
  - Progressive & GEICO have introduced telematics-driven pricing (e.g., Progressive's Snapshot and GEICO's DriveEasy).
  - These use smartphone apps or car sensors to track driving behavior and adjust premiums in real time, competing with Root Insurance's model.
- AI-streamlined Application and Claims Processing
  - Allstate, State Farm, and Liberty Mutual now use AI-driven fully digital onboarding & claims platforms to speed up payouts, similar to Lemonade's instant claims AI.
  - Incumbents integrate AI chatbots + human agents, keeping customer trust while improving efficiency.

Incumbent Responses

2. Acquiring or Partnering with InsurTechs

- Acquisitions of InsurTechs
  - Allstate acquired SquareTrade (a digital warranty and device protection company).
  - Munich Re acquired relayr, an IoT-driven insurance technology firm.
- Partnerships with Startups
  - Chubb partnered with insurtech startup Cover Genius to offer embedded insurance in e-commerce.
  - AXA collaborates with Lemonade's technology in some markets to improve underwriting.

Incumbent Responses

3. Offering More Flexible, On-Demand, and Embedded Insurance

- Launching On-Demand & Usage-Based Insurance
  - GEICO & Allstate now offer similar pay-as-you-go policies to compete.
  - Allianz & Chubb embed travel insurance at checkout for airline tickets.
- Partnering with Tech & E-Commerce Giants
  - Allstate & Amazon partnered to offer insurance bundled with smart home devices.
  - AXA & Tesla partnered to offer embedded auto insurance for Tesla drivers.

Incumbent Responses

4. Leveraging Their Brand Trust & Scale

- Emphasizing Financial Strength & Stability
  - Traditional insurers highlight their decades of experience and strong balance sheets to contrast with loss-making startups.
  - Example: Lemonade and Root Insurance have struggled with profitability, while Allstate & State Farm remain financially stable.
- Using Large Customer Bases for Competitive Advantage
  - Existing insurers cross-sell bundled policies (home, auto, life) at discounts, making it harder for startups to compete on price.
  - Progressive & State Farm leverage massive marketing budgets to maintain strong customer retention.